AMRITA SCHOOL OF ENGINEERING, BENGALURU

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

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23ECE184 INTRODUCTION TO INTERNET OF THING LAB

SYSTEM DESIGN REPORT

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Submitted by

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Smart Parking Detection System

1. Objectives

- To detect real-time vehicle occupancy across three parking slots using IR and ultrasonic sensors.
- To implement dual-sensor validation per slot to eliminate false positives in vehicle detection.
- To control Red/Green LED indicators for each parking slot to provide local physical status indication.
- To transmit live slot status wirelessly to a mobile application using the ESP32 built-in Wi-Fi module.
- To develop a four-screen mobile application using MIT App Inventor for complete parking management.
- To adaptively update slot availability on the app every 3 seconds by polling the ESP32 HTTP server.
- To allow users to select and confirm a booking for an available parking slot through the mobile interface.
- To implement a live parking duration timer and calculate cost automatically at Rs. 50 per hour.
- To generate a detailed receipt showing slot name, entry time, total duration, and total amount on exit

2. Components Required

Components	*Specification*	*Qty*	*Purpose*
ESP32 Dev Board	Dual-core, built-in WiFi/BT, 3.3V logic, 30+ GPIO	1	Main controller + WiFi HTTP server
IR Sensor Module	Digital OUT, 3.3V supply, adjustable sensitivity	3	Primary vehicle detection per slot
HC-SR04 Ultrasonic	2cm–400cm range, 5V supply, 15ms cycle time	3	Distance confirmation, eliminates false IR triggers
Red LED 5mm	VF = 2.0V, IF = 20mA, wavelength 620–625nm	3	Slot occupied indicator
Green LED 5mm	VF = 2.1V, IF = 20mA, wavelength 520–525nm	3	Slot available indicator
220Ω Resistor	1/4W carbon film, 5% tolerance	6	Current limiting for LEDs
Breadboard 830-point	Full-size tie-point breadboard	1	Component mounting and prototyping
M-M Jumper Wires	20cm, various colours	30+	Internal breadboard connections
M-F Jumper Wires	20cm, various colours	20+	Sensor module to breadboard/ESP32
USB Cable	Micro/Type-C matching ESP32 board	1	Programming and power supply
Android Smartphone	Android 5.0+, Wi-Fi capable	1	MIT App Inventor interface

Table 1: Complete list of components used in the Smart Parking Detection System.

3. Block Diagram

The system block diagram illustrates the signal flow from the sensor layer through the ESP32 controller to the WiFi HTTP server, which the MIT App polls every 3 seconds for live slot status updates.

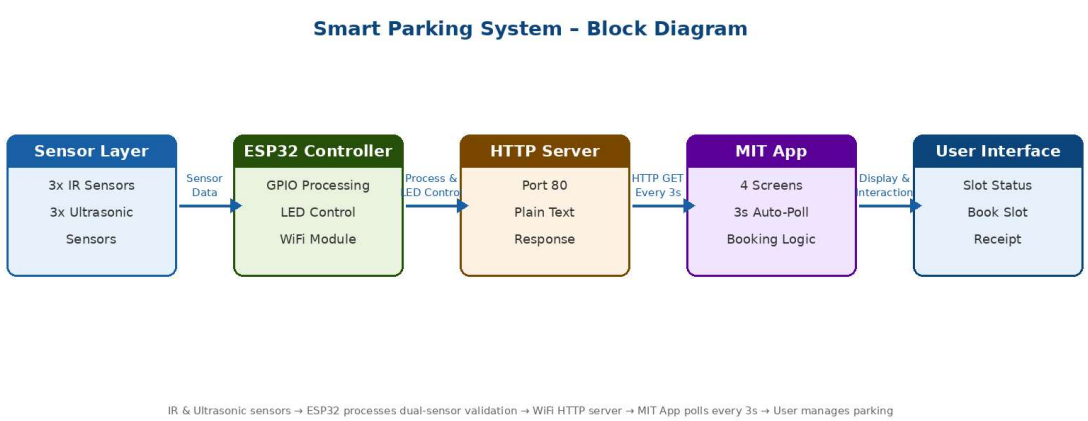


Figure 1: System block diagram showing the complete signal flow. IR and ultrasonic sensors feed into the ESP32, which processes data, controls LEDs, and serves slot status over WiFi. The MIT App fetches and displays live status every 3 seconds.

The block diagram shows four major layers: (1) Sensor Layer with 3 IR sensors and 3 HC-SR04 ultrasonic sensors providing dual-sensor detection per slot; (2) ESP32 Controller processing all sensor data and controlling 6 LEDs; (3) Wi-Fi HTTP Server hosted on the ESP32 serving slot status as plain text; and (4) MIT App Inventor Dashboard displaying live slot availability and managing booking, timing, and cost calculation.

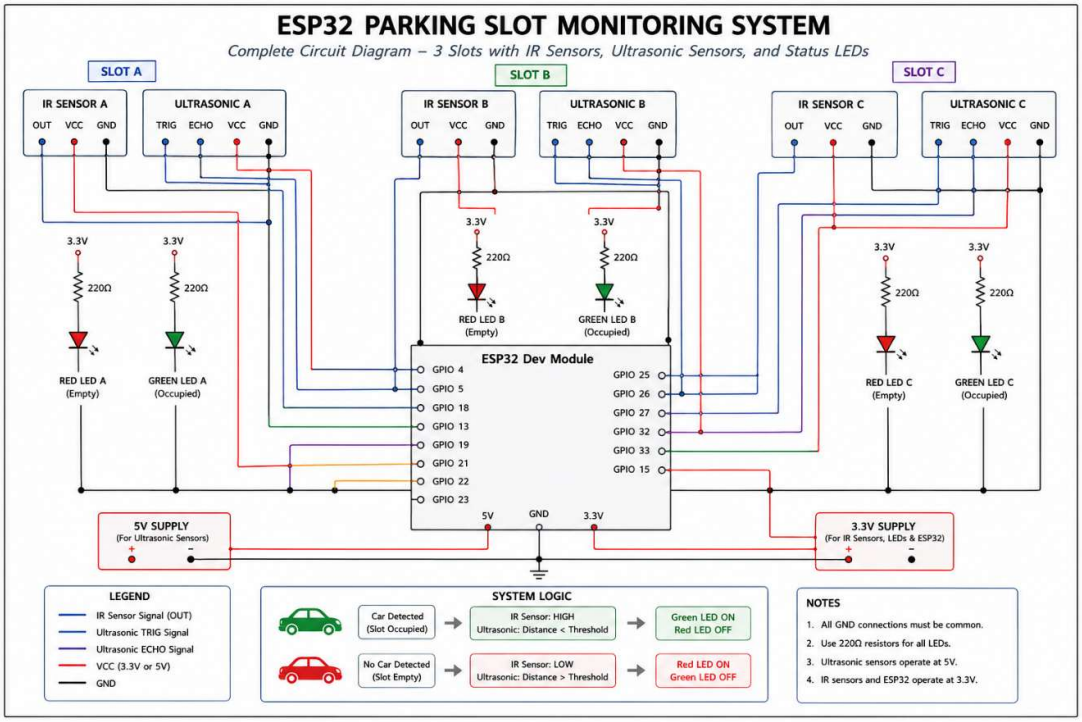
4. Circuit Diagram

4a. Pin Connection Table

Component	***Pin***	***ESP32 GPIO / Rail***
IR Sensor A	OUT / VCC / GND	GPIO 4 / 3.3V Rail / GND Rail
IR Sensor B	OUT / VCC / GND	GPIO 13 / 3.3V Rail / GND Rail
IR Sensor C	OUT / VCC / GND	GPIO 14 / 3.3V Rail / GND Rail
Ultrasonic A	TRIG / ECHO / VCC / GND	GPIO 5 / GPIO 18 / VIN 5V / GND
Ultrasonic B	TRIG / ECHO / VCC / GND	GPIO 19 / GPIO 21 / VIN 5V / GND
Ultrasonic C	TRIG / ECHO / VCC / GND	GPIO 22 / GPIO 23 / VIN 5V / GND
Slot A Red LED	Anode → 220Ω → GPIO / Cathode → GND	GPIO 25 / GND Rail
Slot A Green LED	Anode → 220Ω → GPIO / Cathode → GND	GPIO 26 / GND Rail
Slot B Red LED	Anode → 220Ω → GPIO / Cathode → GND	GPIO 27 / GND Rail
Slot B Green LED	Anode → 220Ω → GPIO / Cathode → GND	GPIO 32 / GND Rail
Slot C Red LED	Anode → 220Ω → GPIO / Cathode → GND	GPIO 33 / GND Rail
Slot C Green LED	Anode → 220Ω → GPIO / Cathode → GND	GPIO 15 / GND Rail

Table 2 / Figure 2: Complete pin connection table. IR sensor VCC pins connect to 3.3V rail. Ultrasonic VCC pins connect to 5V VIN rail. Each LED anode connects through a 220Ω resistor to its GPIO pin. All LED cathodes and sensor GND pins share the breadboard GND rail.

4b. Circuit Diagram Image



[Figure 2 — Circuit Diagram: Full breadboard wiring with ESP32, 3 IR sensors, 3 HC-SR04 sensors, and 6 LEDs with resistors — Attach image here]

Figure 2: Complete circuit diagram showing breadboard layout. The ESP32 is connected via Male-to-Female jumper wires. IR sensors use 3.3V supply. Ultrasonic sensors use 5V VIN supply. All LEDs are connected through 220Ω current-limiting resistors. All GND connections share the common breadboard ground rail.

5. Working Prototype

The working prototype consists of an ESP32 Dev Board connected via Male-to-Female jumper wires to three IR sensor modules, three HC-SR04 ultrasonic sensors, and six LED indicators (three red, three green) on a breadboard. The system continuously reads all sensors and updates LED status in real time while serving slot data over WiFi.

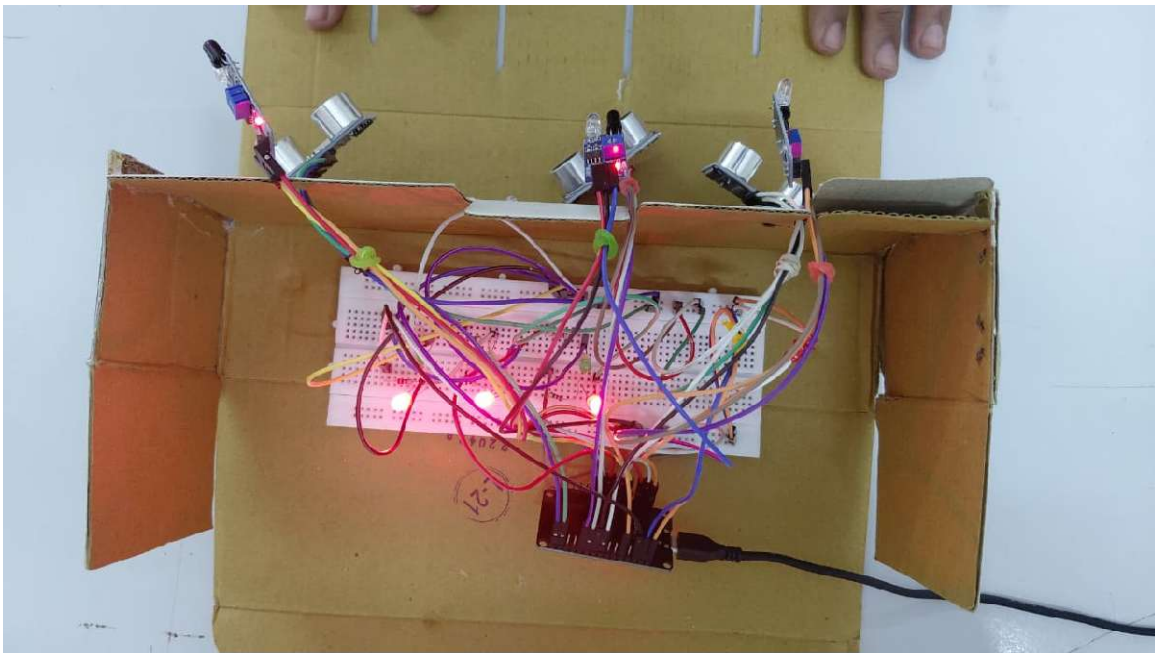


Figure: Top View

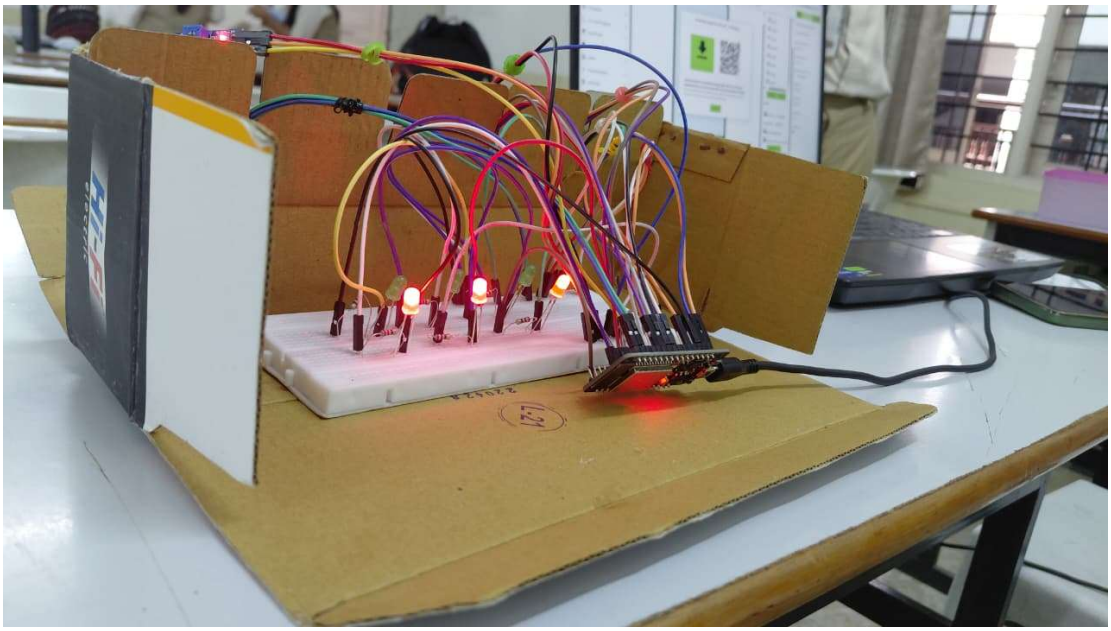


Figure: Side View

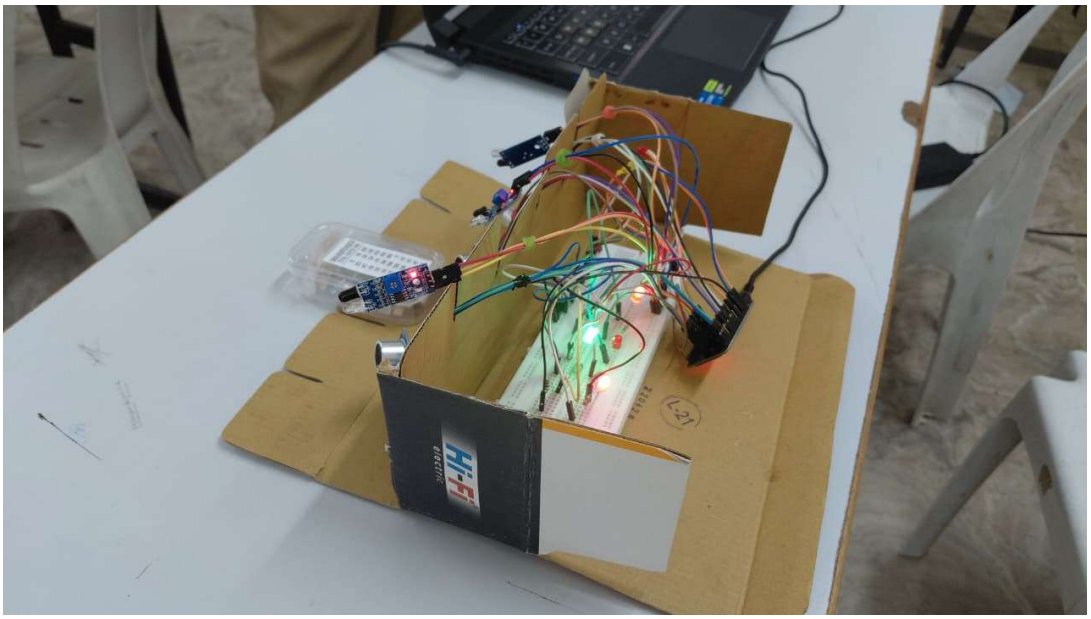


Figure: Working View (Slot – B Occupied)

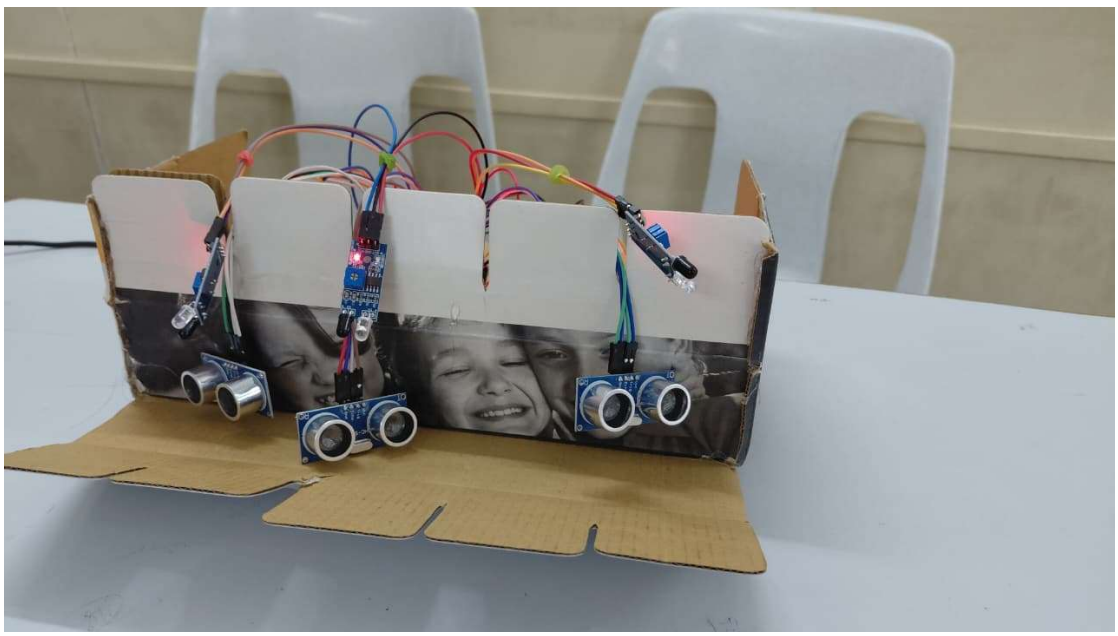
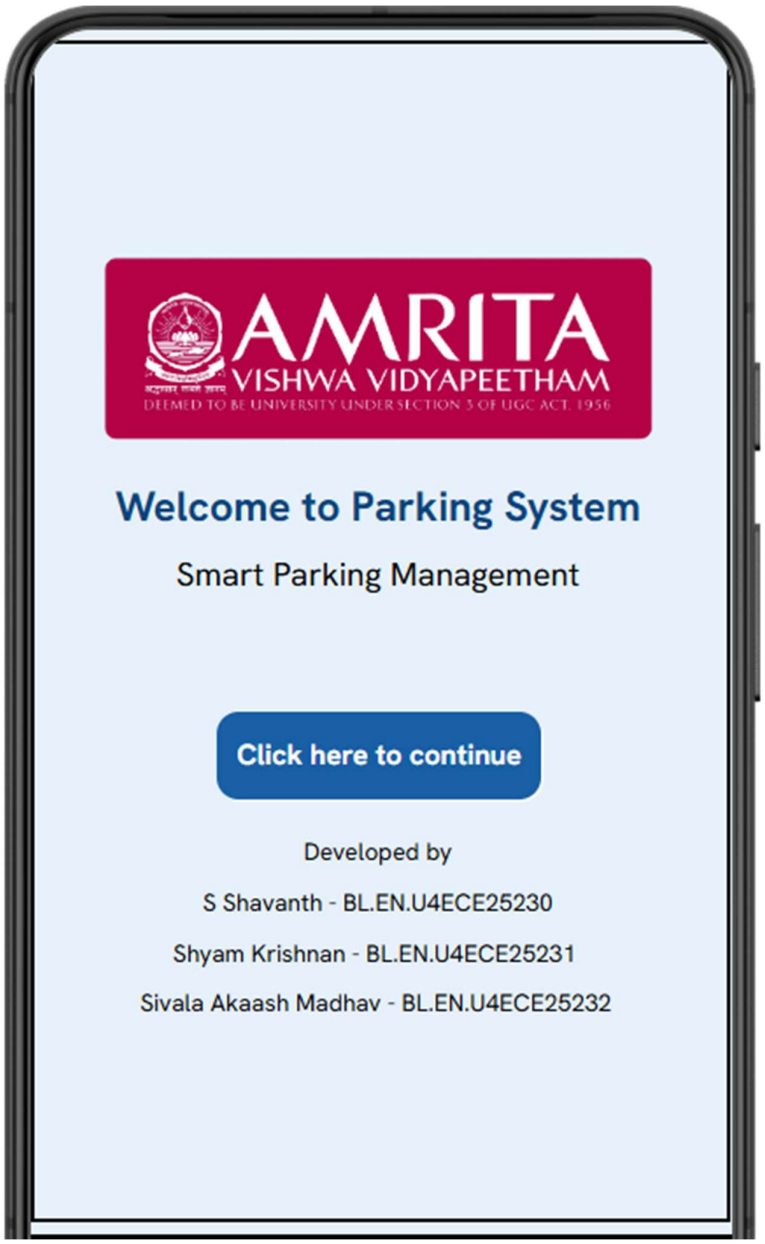


Figure: Front View

6. MIT App Screenshots

The MIT App Inventor application consists of four screens designed for complete parking management — from slot selection through booking confirmation, live session tracking, and payment receipt generation.

Figure 4a: Welcome Screen (Screen 1)



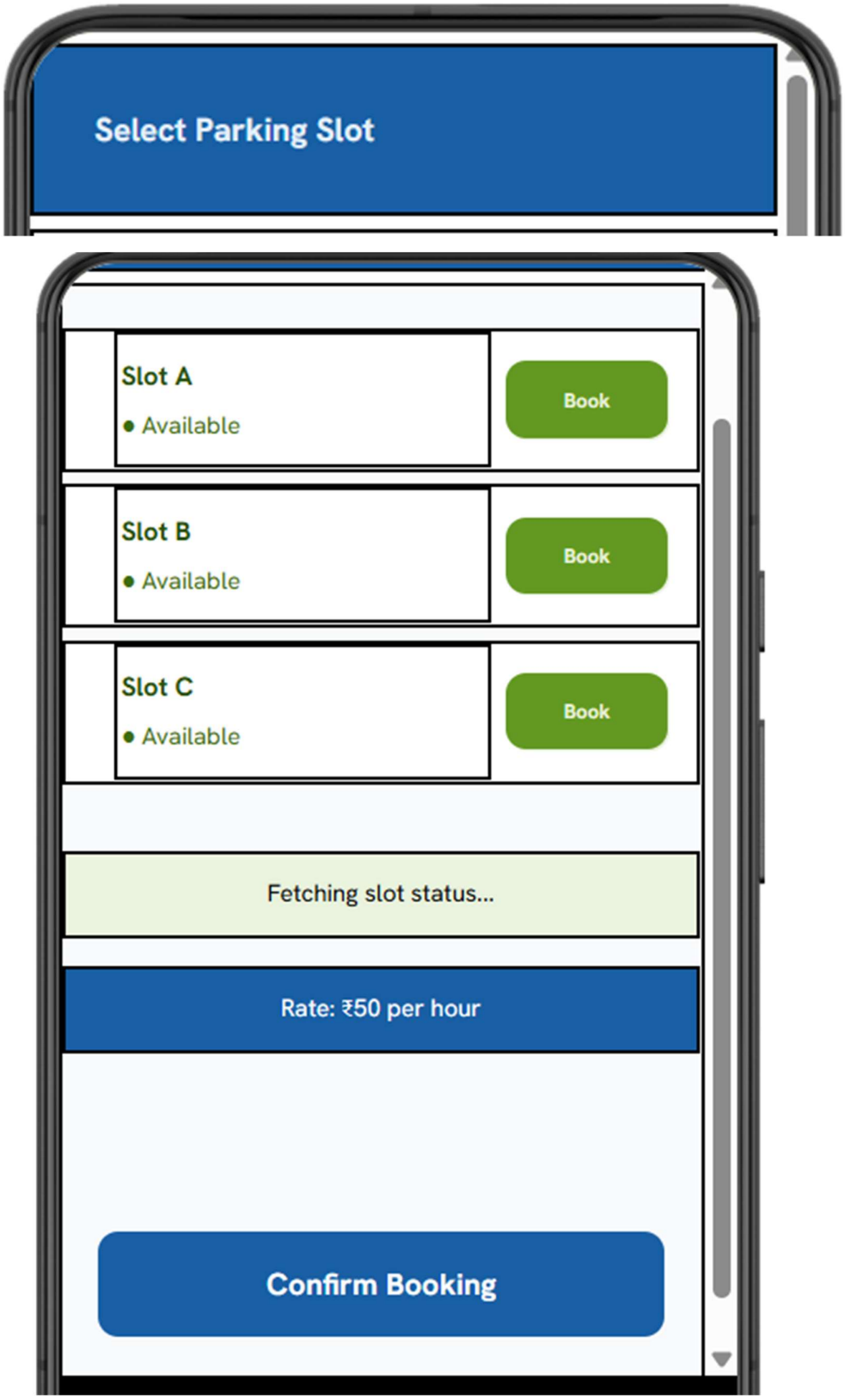


Figure 4b: Slot Selection Screen (Screen 2)

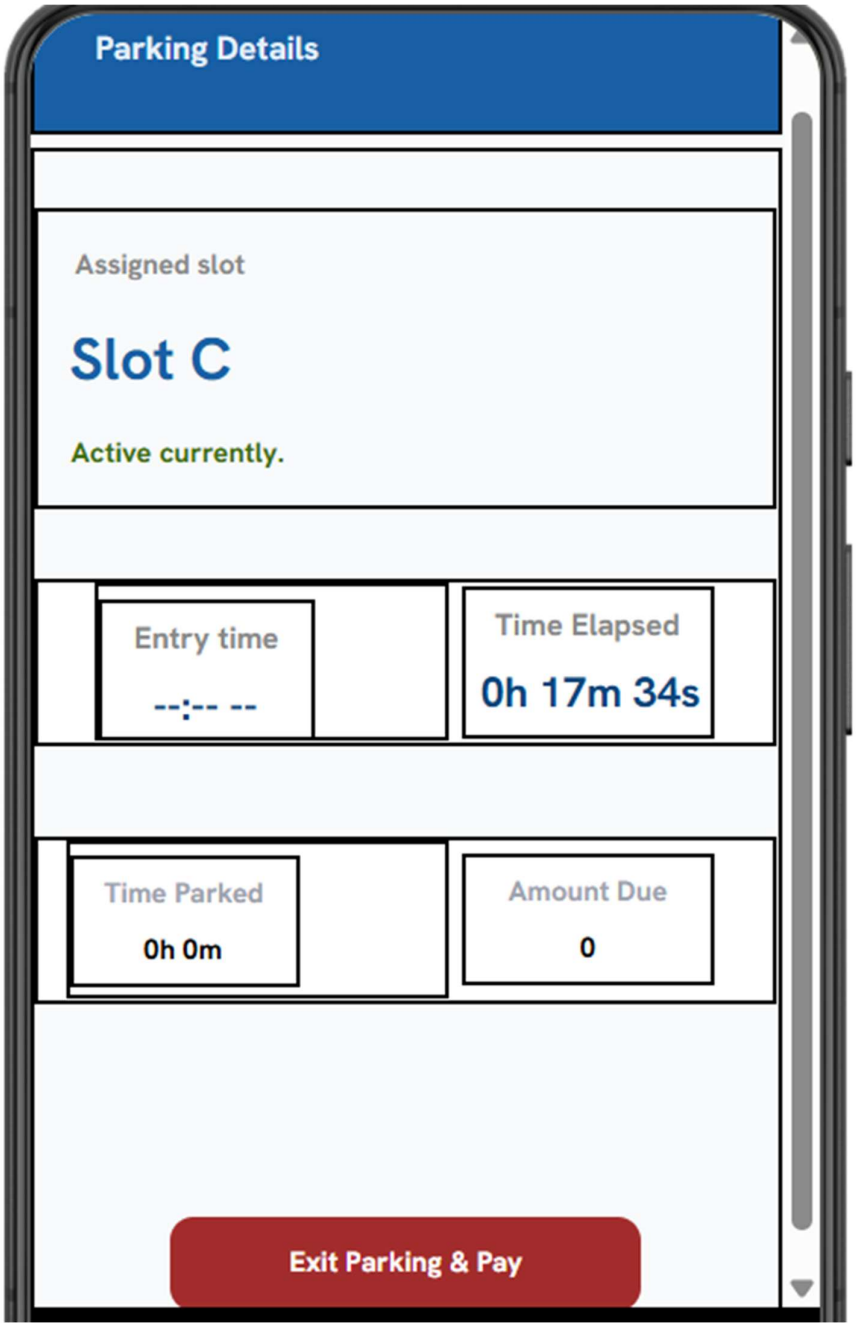


Figure 4c: Parking Details Screen (Screen 3)

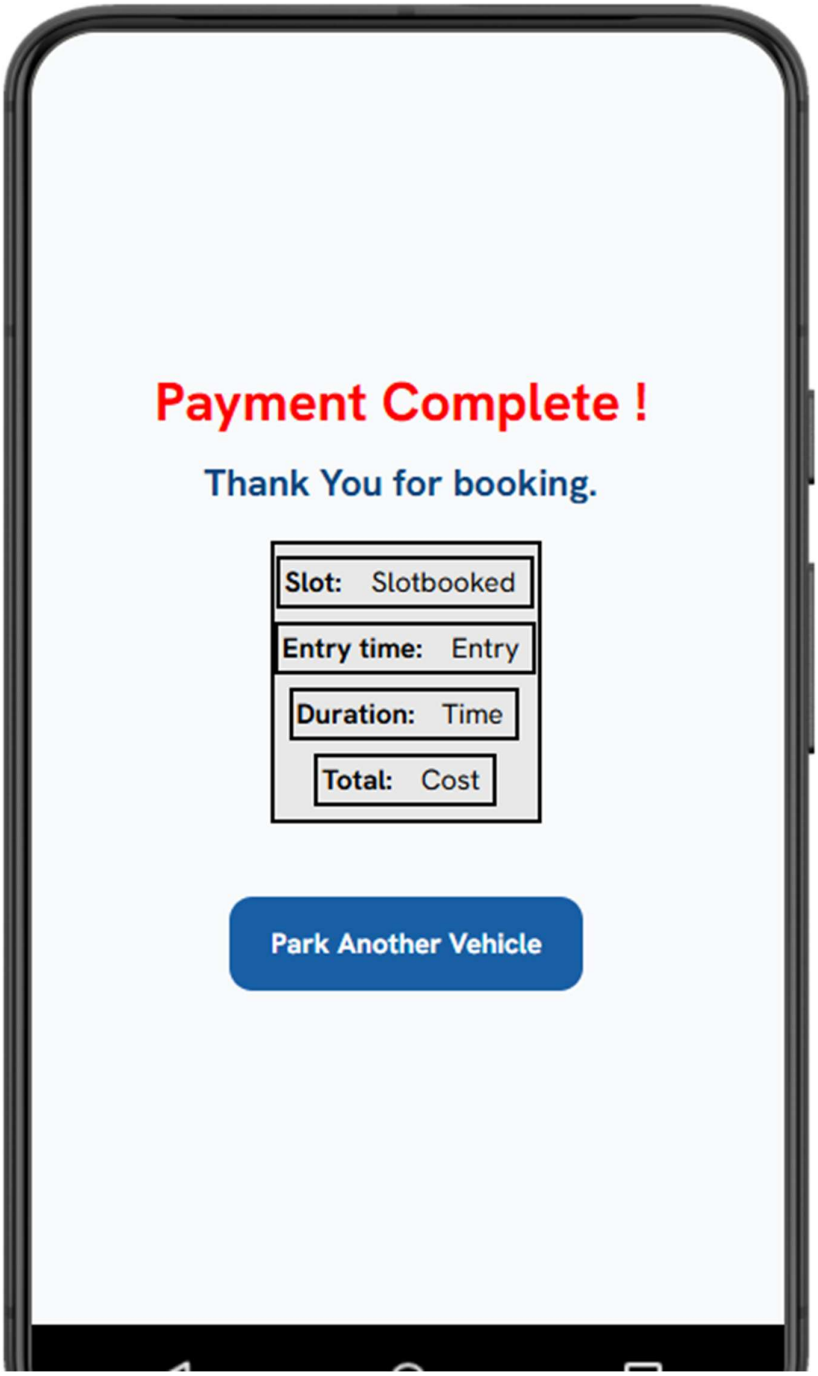
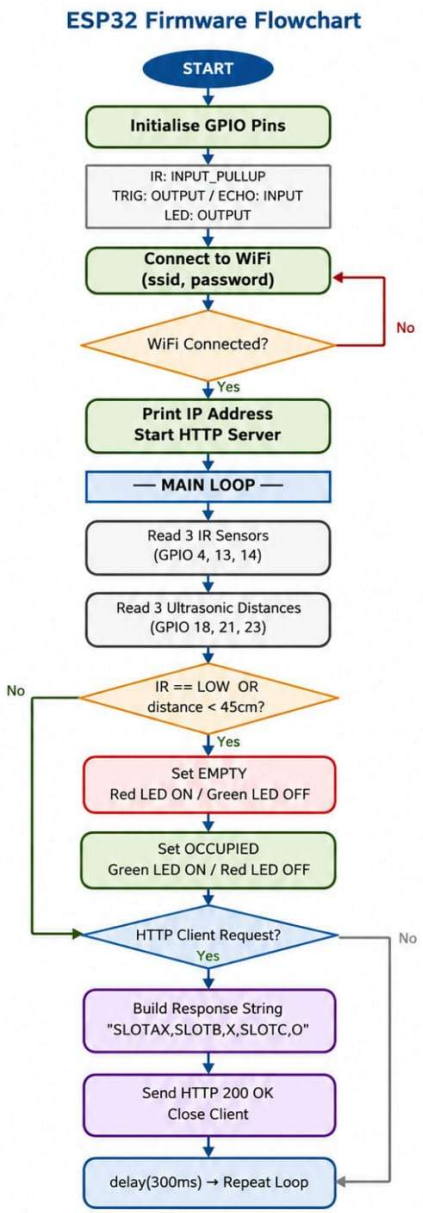


Figure 4d: Receipt Screen (Screen 4)

7. Flow Charts

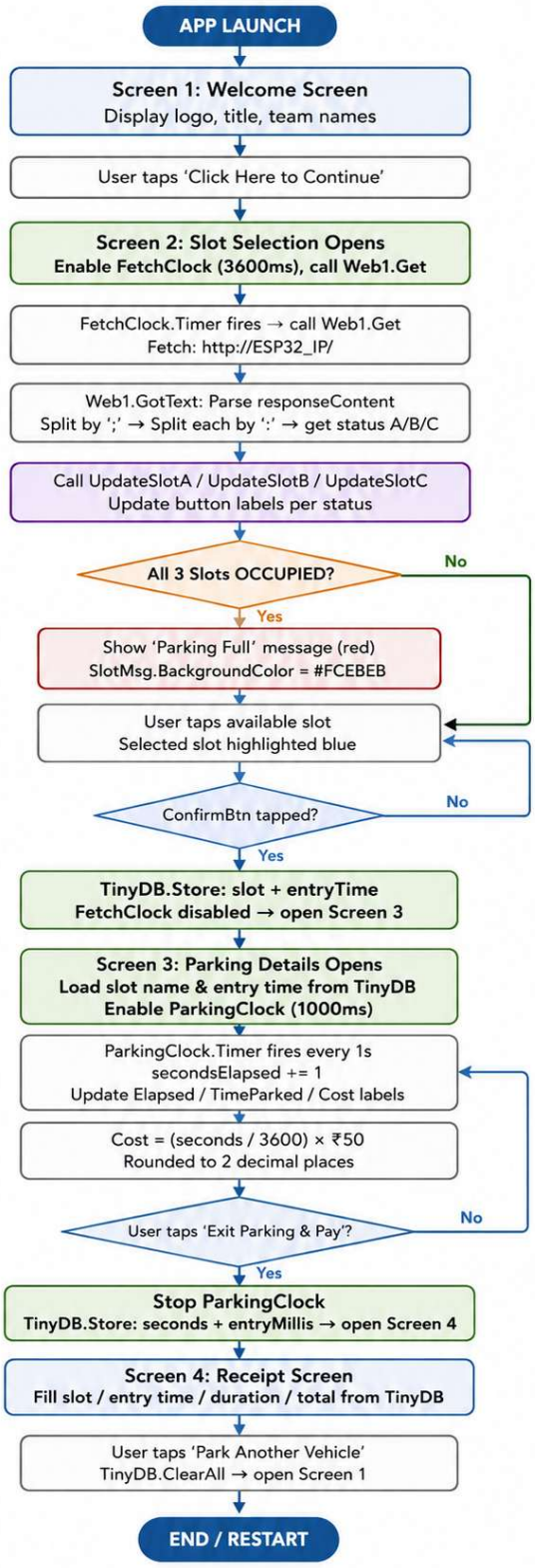
Flow charts for both the ESP32 firmware and the MIT App Inventor application were generated using draw.io. Hand-drawn charts are not used as per the project guidelines.

7a. ESP32 Firmware Flow Chart



7b. MIT App Inventor Application Flow Chart

MIT App Inventor – Application Flowchart



8. GitHub Link

The complete ESP32 firmware source code for this project has been uploaded to GitHub. The repository contains the Arduino file with all sensor reading functions, dual-sensor validation logic (IR + ultrasonic), LED control, WIFI connection with reconnection handling, and HTTP server response code.

Repository Link	https://github.com/carnotengine-coal/Smart-parking-system-esp32-.git

9. References

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